Abhishek Bose

(+91)9163309846 abhishekabcbose@gmail.com Portfolio
LeetCode LinkedIn GitHub

Education

Techno India University, Kolkata Aug. 2024 – Jul. 2026
Master of Computer Applications (MCA) *CGPA: 8.13/10*

Techno Main Salt Lake, Kolkata Aug. 2021 – Jul. 2024
Bachelor of Computer Applications (BCA) *CGPA: 7.60/10*

Experience

Zidio Development | *Web Development Intern* Jun. 2024 – Aug. 2024

- Built a website where people can find jobs and create resumes, using React.js, Node.js, Express.js and MongoDB, with REST APIs (the connections that let the pages and the server talk to each other).
- Let users download their resumes as PDF or PNG files, created in the background so the app never freezes; helped connect APIs, fix bugs and make the backend faster, using Git to collaborate with the team.

Projects

Collaborative Code Editor | *TypeScript, Yjs, Socket.IO, Docker, AWS ECS Fargate* GitHub

- Packaged the app into a portable container using Docker, stored it in Amazon's online storage (ECR), and ran it on Amazon's cloud (ECS Fargate) without managing any servers.
- Built a real-time collaborative code editor with React, TypeScript, HTML, CSS and JavaScript, where multiple people can edit the same code at once. Used Yjs (a tool that merges simultaneous edits without conflicts) and Socket.IO (for instant live syncing).

HR Management System | *Odoo, Docker, Git* GitHub | Hackathon Finalist

- Hackathon finalist (Odoo × Adamas, 1,600+ teams); led backend development for an HR management system that handles employee records and workflows.
- Built the backend in TypeScript, storing data in a MySQL database and using Prisma (a tool that lets the code talk to the database easily); worked in an agile environment with short build-and-review cycles.

Vector Database & RAG Pipeline | *Java, REST API, Ollama* GitHub

- Built a Java database that finds "things that look alike" using three search methods (HNSW, KD-Tree, brute-force) and three ways of measuring closeness (cosine, Euclidean, Manhattan).
- Created REST APIs (the connections other apps use to talk to it) for storing, searching and testing vectors; added Ollama embeddings and LLM-based RAG so people can ask questions in plain English and get answers from their documents.

UPI Without Internet | *Java, Spring Boot, Distributed Systems* GitHub

- Designed a device-to-device payment system for offline transfers using Spring Boot, hybrid encryption, and idempotent message processing to prevent duplicate payments during retries.
- Executed transaction correctness, secure communication, and failure handling under intermittent connectivity.

Open-Source Contributions

Apache Fineract Consumer Facing | *Java, Spring Boot, Angular, TypeScript, Playwright* PR #196 | PR #205

- Gave the code that handles changes (like creating or updating records) clearer names that follow a standard industry pattern (CQRS), across the login and beneficiary parts of the app.
- Acted on feedback from the project's maintainers: removed a list of known exceptions to the automatic architecture checks (ArchUnit) and updated the architecture documentation. The change was merged.
- Reworked the Angular and TypeScript frontend so the language-translation feature (I18nService) and the device-identification feature (DeviceFingerprint) use shared, standard connectors (I18N and STORAGE adapters), keeping both in one consistent place.
- Updated every file and test that used these features, and passed the project's automated CI/CD checks, including linting (automatic code-style checks) and Playwright tests (automatic tests that click through the app like a real user), before the maintainers merged it.

Skills

Languages: Java, JavaScript, Typescript, SQL
Java & Backend: Java, OOP, Collections, Exception Handling, REST APIs, Spring Boot, Spring Data JPA
Databases: MySQL, MongoDB, Relational Databases
Backend Concepts: Authentication, Idempotency, Distributed Systems, RAG Pipelines
Tools & Practices: Docker, Git, GitHub, GitHub Actions, Vercel, Render, Agile Vercel, Render
Frontend: React.js, TypeScript, Zustand, Redux, Tailwind CSS
Core CS: DSA, Hashing, File I/O, Object-Oriented Design